



**PCMR EM
SCHOOL**

SUMMATIVE ASSESSMENT - 3

Name :

Class : Jr. 10TH

Section : ALL

Subject : MATHEMATICS

Marks : 100

Time : 3 Hours



I Answer any three Questions

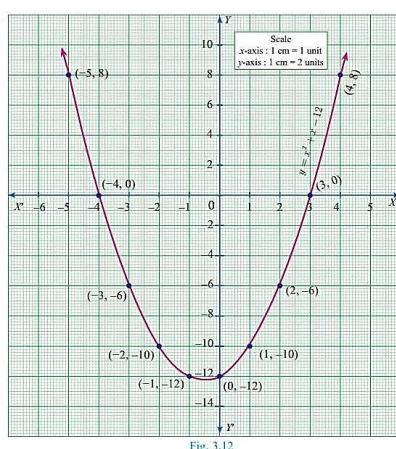
3×8 = 24m

1. Prove that $\sqrt{5}$ is irrational.
2. Prove that $3+2\sqrt{5}$ is irrational.
3. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting:
 - i) a king of red colour
 - ii) a face card
 - iii) a spade
 - iv) the queen of diamonds
4. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, Find the probability that it bears:
 - i) a two-digit number
 - ii) a perfect square number
 - iii) a number divisible by 5.

II Answer all the following Questions.

3×4 = 12m

5. Suppose we throw a dice once. "What is the probability for getting a number greater than 4"? Now create 4 such type of questions.
6. Find the zeroes of the following quadratic polynomial and verify the relationship between the zeroes and the coefficients: $x^2 - 2x - 8$
7. By observing the graph answer the following questions.



- i) What is the shape of the graph.
- ii) What is the name of polynomial in the graph.
- iii) Find the sum of the zeros.
- iv) Find the product of the zeros.

4x2 = 8m

- IV Answer all the questions in one word or phrase.**

6x1 = 6m

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C) 4 D) 5

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C) 6 D) 9

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D) 0.8

All the Best....